Coordinate Geometry

Section 1: Lines, Lengths & Midpoints

1 Find the equation of the line passing through $A(2, -3)$ and $B(6, 5)$ in the form $y = mx + c$.

a. $y = 2x - 7$

b. $y = 2x + 1$

c. $y = \frac{1}{2}x - 4$

d. $y = 2x - 3$

2 Find the length of the line segment joining $P(-1, 4)$ and $Q(5, -4)$.

a. 10

b. 8

c. $2\sqrt{7}$

d. 12

3 Find the midpoint of the segment connecting $C(-3, 8)$ and $D(7, -2)$.

a. $(2, 3)$

b. $(5, 3)$

c. $(2, 5)$

d. $(4, 6)$

Section 2: Perpendicular Bisectors & Gradients

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4 Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(5, 10)$.

a. $x + 2y = 15$

b. $2x - y = 0$

c. $x - 2y = -9$

d. $2x + y = 12$

5 The line L_1 has equation $3x - 4y = 12$. Find the equation of the line L_2 perpendicular to L_1 passing through $(2, 1)$.

a. $4x + 3y = 11$

b. $3x + 4y = 10$

c. $4x - 3y = 5$

d. $3x - 4y = 2$

6 Three points are given: $P(1, 1)$, $Q(4, 5)$, and $R(k, 9)$. If P, Q, R are collinear, find k.

a. 7

b. 6

c. 8

d. 5

Section 3: Circles — Equations, Tangents & Normals

7 Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y - 11 = 0$.

a. Center $(3, -4)$, radius 6

b. Center $(-3, 4)$, radius 6

c. Center $(3, -4)$, radius $\sqrt{11}$

d. Center $(-3, 4)$, radius 36

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8 Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

a. $3x + 4y = 25$

b. $4x - 3y = 0$

c. $3x - 4y = 25$

d. $4x + 3y = 24$

9 A circle has diameter with endpoints $A(-2, 1)$ and $B(4, 9)$. Find the equation of the circle.

a. $(x - 1)^2 + (y - 5)^2 = 25$

b. $(x + 1)^2 + (y + 5)^2 = 25$

c. $(x - 1)^2 + (y - 5)^2 = 100$

d. $(x - 3)^2 + (y - 4)^2 = 25$

Section 4: Intersections & Exam-Standard Geometry

10 Determine the relationship between the line $y = x - 5$ and the circle $x^2 + y^2 = 9$.

a. The line does not intersect the circle

b. The line is tangent to the circle

c. The line intersects the circle in two distinct points

d. The line passes through the center of the circle

11 Find the area of the triangle with vertices $A(0, 0)$, $B(6, 2)$, and $C(2, 8)$.

a. 22

b. 20

c. 24

d. 18

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12 Find the length of the chord cut off from the line $y = 2$ by the circle $x^2 + y^2 = 13$.

a. 6

b. 3

c. $2\sqrt{13}$

d. 8

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